Preregistered Rejection as a First-Class Negative Result

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1 Preregistered Rejection as a First-Class Negative Result

A methodological note on pre-execution refusal in preregistered research

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1.1 Abstract

Pre-registration has become a standard practice for reducing researcher degrees of freedom, and negative results from preregistered protocols are increasingly valued. But preregistration discourse is asymmetric: it tells researchers *when* to run an experiment and *how* to interpret its result, but is silent on *whether* to run it at all. In particular, when an analyst realizes — before execution — that the proposed consequence measure is analytically coupled to the structural variable it is supposed to measure independently, the standard workflow offers no formal category for this refusal.

We propose **preregistered rejection** as a third category of preregistered output, alongside preregistered *success* and preregistered *failure*. A preregistered rejection consists of (i) a proposed experiment with its full four-tuple specification, (ii) a *pre-execution* decision to refuse, (iii) an analytic or empirical justification that the refusal can be verified independently of the analyst's later beliefs, and (iv) a set of *unexplored alternatives* the analyst considered but declined to pursue.

We illustrate the category with a case from our own project: an intervention study (PR-2) that FAILed at three sample sizes, motivated by its own diagnosis, proposed a substitution (PR-2b) — replacing `drop_magnitude` with `recovery_AUC` as the consequence variable. A pre-execution analysis showed that `modulator_Ra = 1 − bridge_fraction − 0.5·ap_fraction` and `recovery_AUC = mean_t(r(t)) / drop_magnitude` share a component: `bridge_fraction` appears in the modulator, and `drop_magnitude` is the normalizer. Because deleting a bridge edge decreases both the numerator and the denominator of `bridge_fraction`, the modulator's penalty decreases, R-a inflates, and `recovery_AUC` inflates — the intervention that "broke" the system is *predicted by construction* to raise the proposed recovery metric. The pre-execution analysis recorded an artifact rate of 95.2% on the intervention arm with 85.7% reverse separation on the control arm, and a merged-arm correlation |corr| = 0.902. The proposed experiment was not run.

We argue that preregistered rejection is a genuine scientific contribution — not a "failed attempt" but a distinct category of negative result whose value is comparable to preregistered failure. We formalize the category, describe its machine-enforcement (Gate 47 in our project), and discuss the substrate limits it reveals: when no independent second read-out exists in the current metric family, the boundary is a **scope boundary** — a call for a new instrument — not a sample-size failure.

Keywords: preregistration, negative results, pre-execution refusal, measurement redundancy, causal inference, self-verification, computational experiments

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1.2 1. Introduction

1.2.1 1.1 The asymmetry in preregistration

Pre-registration has been adopted as a standard practice to reduce researcher degrees of freedom and to protect the interpretive value of null and negative findings. It has generated a rich literature on:

- *When* to run an experiment (e.g., power calculations, sample size justification);
- *How* to specify success and failure criteria in advance (e.g., preregistered thresholds, equivalence tests);
- *How* to interpret negative results (e.g., equivalence vs. absence-of-evidence; replication registries).

But preregistration discourse is silent on a third question: **whether to run the experiment at all**, when pre-execution analysis reveals that the proposed measurement is analytically redundant with the intervention. This is a distinct case from underpowered studies, under-specified hypotheses, or ill-designed controls: the failure mode is not statistical, it is *definitional*.

1.2.2 1.2 The gap

Consider a scenario common in computational and networked research:

1. An initial experiment (E1) is preregistered and FAILS at three sample sizes.
2. E1's own diagnostic points to a candidate substitute measurement M'.
3. E1's FAIL is thus reframed: perhaps E1 was measuring the wrong thing, and re-running with M' would succeed.
4. The analyst proposes E2 = E1 with M' as the outcome variable.

The standard workflow has no formal category for E2 being *refused* before execution. The analyst is effectively forced to choose between two dishonest options:

- **Run E2 anyway**, publish "another failure," and let the pattern itself be discovered by readers; or
- **Refuse E2 without documentation**, effectively hiding an important methodological decision.

Both options hide the *decision process* from future readers. The first hides it by noise; the second hides it by silence.

1.2.3 1.3 Our proposal

We propose **preregistered rejection** as a formal category for this third case. A preregistered rejection records:

- The proposed experiment (E2) with its full preregistered specification;

- The pre-execution decision to refuse;
- A justification that is verifiable independently of the analyst's later beliefs;
- A set of *unexplored alternatives* the analyst considered but declined to pursue;
- An explicit statement of what condition would need to be satisfied to lift the refusal.

This category sits alongside preregistered success and preregistered failure as a **third class of preregistered output**, with its own interpretive value.

1.2.4 1.4 Roadmap

Section 2 formalizes the category. Section 3 introduces Gate 47, our machine-enforced integrity checker. Section 4 presents the case study. Section 5 discusses the implications for preregistration discourse and the substrate limits this case reveals. Section 6 concludes.

1.3 2. Formalization: preregistered rejection

1.3.1 2.1 Definition

Definition 1 (Preregistered experiment). A *preregistered experiment* is a tuple

$$E = \langle S, I, M, C, \Theta \rangle$$

where S is the target system, I is the intervention (a set of structural actions), M is the consequence measure, C is the control, and Θ is the tuple of preregistered success criteria.

Definition 2 (Preregistered rejection). A *preregistered rejection* of E is a tuple

$$R_E = \langle E, D, J, A, U, T \rangle$$

where:

- $D = \texttt{REJECT}$ is the pre-execution decision;
- $J = (J_{\text{analytic}}, J_{\text{empirical}})$ is a justification that is (a) analytically derivable from the definitions of M and I alone, or (b) empirically detectable on a sample that is independent of E's intended sample;
- $A = \texttt{criteria_modified_after_results} = \texttt{False}$ is an explicit declaration that Θ was not revised in response to R_E's evidence;
- U is the set of *unexplored alternatives* — substitutes, follow-ups, and scope changes the analyst considered but declined to pursue;
- T is a *lift condition* — the condition under which R_E would be lifted and E reconsidered.

1.3.2 2.2 Independence of justification

A critical property of a preregistered rejection is that its justification must be **independent of the analyst's later beliefs**. If J requires the analyst to know the outcome of E in order to justify R_E, then R_E is not a refusal — it is post-hoc rationalization.

Two forms of independence:

- **Analytic independence**: J_{analytic} is derivable by reading the definitions of M and I, without any data. In this case, refusal is unambiguous: the analyst can prove that running E is definitionally redundant before any data is collected.
- **Empirical independence**: $J_{\text{empirical}}$ uses a sample that is *provably independent* of E's intended sample. In this case, refusal is justified only if the sample was selected before the analyst knew whether E would succeed; if the sample is drawn after E is proposed but before E is executed, the analyst must document the selection rule and show that it does not depend on E's predicted outcome.

1.3.3 2.3 The unexplored alternatives field

A preregistered rejection is *incomplete* without its U field. The U field has three purposes:

1. **Reproducibility**: future readers can reconstruct not just what the analyst did but what she considered and declined;
2. **Auditability**: an R_E with an empty U field is a red flag — it suggests the analyst did not actually consider alternatives;
3. **Boundary communication**: the U field implicitly specifies the scope boundary of the current measurement family, i.e., what conditions would need to change for the refusal to be lifted.

1.3.4 2.4 Interpretive value

We claim that preregistered rejection has **comparable interpretive value** to preregistered success and failure. Its value lies in three places:

1. **Documentation of decisions not taken** — every refusal has a cost; documenting it prevents silent abandonment of a methodological branch;
2. **Boundary identification** — a preregistered rejection pinpoints the substrate limits of the current measurement family, which is itself a scientific finding;
3. **Anti-rationalization** — by requiring U and T to be documented *before* execution, the format prevents post-hoc rationalization of why the experiment was not run.

1.4 3. Machine enforcement: Gate 47

1.4.1 3.1 Rationale

A preregistered rejection is a *claim about a decision not taken*. Such a claim has no empirical data on which to fall back: it must be verified through its own specification. This makes it vulnerable to a class of abuse — post-hoc rationalization — that preregistered success and failure are immune to (they have data to check against).

We therefore introduced **Gate 47** — a machine-enforced integrity checker that verifies:

- The preregistered rejection's specification matches the executed pre-execution analysis;
- The `criteria_modified_after_results` field is `False`;
- The `unexplored_alternatives` field is populated and non-trivial;
- The decision is derived from an independently-computable quantity (not the analyst's summary field);
- The rule used to make the decision is itself under-documented (i.e., `rule_underpowered = True` is explicitly recorded if applicable).

1.4.2 3.2 The independent recomputation principle

Gate 47 follows a general principle: **the checker should recompute the decision from the raw sample, not from the analyst's summary**. This means that if the analyst's summary field is tampered with, the checker will catch the mismatch.

Formally: given a preregistered rejection $R_E = \langle E, D, J, A, U, T \rangle$ and a raw sample X:

- The checker recomputes $J' = \texttt{compute}(X, E)$ from the raw sample;
- It compares J' to J;
- If they differ, it fails the integrity check.

This principle — **independent recomputation > analyst trust** — generalizes beyond Gate 47 to all preregistered experiment integrity checks.

1.4.3 3.3 Implementation

Gate 47 is implemented as `scripts/check_metric_substitution.py` in the `flybrain-connectome` project. The pre-execution analysis script is `scripts/f4b_metric_substitution_check.py`. The full protocol is at `docs/PREREGISTRATION_PROTOCOL_v1.md#PR-2b`.

1.5 4. Case study: PR-2b

1.5.1 4.1 Background: PR-2 and its FAIL

In the same project, we ran a preregistered intervention experiment (PR-2) testing the hypothesis: "deleting a bridge edge of a network increases its `drop_magnitude` (a measure of fault isolation)". The experiment was run at three sample sizes — n = 10, 50, 100 topologies — and FAILed at all three: composite pass rates of 38.9%, 71.4%, and 59.1% respectively; effect sizes |mean_delta| ≤ 0.0005.

Diagnosis: PR-2's own result analysis suggested that `drop_magnitude` was saturated at small n (i.e., it hit its upper bound for many topologies), limiting the measurable effect. A natural substitute was `recovery_AUC` — a measure of recovery dynamics — which was already part of the project's metric family (see PR-4, which PASSed).

1.5.2 4.2 The proposed substitute: PR-2b

PR-2b was proposed as: re-run PR-2 with `recovery_AUC` substituting `drop_magnitude` as the consequence variable. The proposition had all the surface qualities of a legitimate preregistration extension: same intervention, same control, updated metric with a clear motivation from the previous experiment's own diagnosis.

1.5.3 4.3 The pre-execution analysis

Before running PR-2b, we performed an analytic and empirical analysis of whether `recovery_AUC` is *independent* of `drop_magnitude`. Reading `compute_ablation` line by line revealed the following definitions:

$$\texttt{modulator_Ra} = 1 - \texttt{bridge_fraction} - 0.5 \cdot \texttt{ap_fraction}$$

$$\texttt{recovery_AUC} = \frac{\text{mean}_t(r(t))}{\texttt{drop_magnitude}}$$

Two observations follow:

Observation 1 (analytic coupling): `modulator_Ra` contains `bridge_fraction` in its numerator (as a subtracted penalty term). When a bridge edge is deleted, `n_edges` and `n_bridges` both decrease by 1, so `bridge_fraction = (B-1)/(E-1) < B/E` — the bridge fraction *decreases*. That lowers `modulator_Ra`'s penalty term, which *raises* `R-a`, which (via the recovery dynamics) *raises* `recovery_AUC`. In other words: the intervention that "broke" the system is *predicted by construction* to *raise* the proposed recovery metric.

Observation 2 (normalizer coupling): `recovery_AUC` is normalised by `drop_magnitude` — the very quantity PR-2 measures. This means `recovery_AUC` and `drop_magnitude` are coupled at the definitional level: they cannot vary independently.

Both observations are derivable from the definitions alone; no empirical data is required.

1.5.4 4.4 The empirical verification

We ran a pre-execution empirical check (`scripts/f4b_metric_substitution_check.py`) on a fresh sample of 30 topologies (independent of PR-2's intended sample) with 50 nodes each. The check computed:

- **Artifact rate on the intervention arm**: 20/21 = **95.2%** — i.e., in 20 of 21 bridge-edge deletions, `recovery_AUC` *rose* after the "damage";
- **Reverse artifact rate on the control arm**: 18/21 = **85.7%** — i.e., in 18 of 21 non-bridge-edge deletions, `recovery_AUC` *fell* after the "control" intervention;
- **Merged-arm correlation**: |corr(Δbridge_fraction, Δrecovery_AUC)| = **0.902**;
- **Single-arm (non-bridge) correlation**: |corr| = **0.906**;
- **Intervention-arm correlation alone**: |corr| = 0.354 (below the pre-registered rejection threshold of 0.5).

1.5.5 4.5 The rule's own insufficiency

The preregistered rejection rule (|corr| > 0.5) flagged only the merged-arm sample. On the intervention arm alone, the variance of Δbridge_fraction was compressed to 4.82×10^{-5} by the intervention itself (all bridges were deleted), so |corr| = 0.354 fell below the rejection threshold — a spurious "ACCEPT" verdict.

The analyst explicitly recorded `rule_underpowered = true` on the intervention arm and did **not** modify the threshold. The rejection stood on:

- The analytic proof (Observations 1 and 2);
- The merged-arm correlation of 0.902 (far above threshold);
- The artifact rate of 95.2% on the intervention arm with 85.7% reverse separation on the control arm.

1.5.6 4.6 The unexplored alternatives

The preregistered rejection record also documented three alternatives that were considered but not pursued:

1. **Larger n**: rejected because changing n affects effect size, not the denominator artifact;
2. **Half-life of recovery (t½)**: rejected because t½ is derived from the same trajectory, so it carries the same coupling;
3. **An independent second read-out under real dynamics**: identified as the *only* path forward, but noted to be outside the project's current compute substrate.

1.5.7 4.7 The boundary conclusion

With `drop_magnitude` effect size ≤ 0.001 at n ≤ 100, and no independent second read-out in the current metric family, **L4 (causal) evidence for the bridge → isolation link is not attainable within the current substrate**.

This is a **scope boundary** — a call for a new instrument — not a sample-size failure. Framing it as the latter would prescribe more runs of the same instrument; framing it as the former prescribes a new measurement. The distinction matters: it changes the follow-up work from "run more" to "build something different."

1.6 5. Discussion

1.6.1 5.1 The interpretive value of preregistered rejection

Preregistered rejection is not merely "another failure." It has a distinct interpretive value:

- **Documentation of decisions not taken** — every refusal has a cost; documenting it prevents silent abandonment;
- **Boundary identification** — a preregistered rejection pinpoints the substrate limits of the current measurement family;
- **Anti-rationalization** — by requiring U and T to be documented before execution, the format prevents post-hoc rationalization.

1.6.2 5.2 Substrate limits vs. sample-size limits

The PR-2b case is a canonical example of a **substrate limit**: a limitation that cannot be resolved by gathering more data, only by changing the instrument. The two framings prescribe different actions:

	Sample-size limit	Substrate limit
Nature	Underpowered study	Missing measurement
Diagnosis	Effect size ≤ detection threshold	No independent read-out exists
Prescription	Run more samples	Build a new instrument
Failure mode	Not enough data	Wrong instrument

A preregistered rejection that identifies a substrate limit is thus scientifically more informative than a preregistered failure at any sample size: it tells future researchers *not* to run more of the

same.

1.6.3 5.3 The R-0.4 extension

The R-0.4 discipline from our project's preregistration protocol states: *"failed experiments are reported, not adjusted."* PR-2b extends this to: *"not-run experiments are also reported, with their reason."* This is a small but important extension:

- **Without R-0.4**: the analyst is free to abandon E2 without documentation.
- **With R-0.4 (extended)**: the analyst must record the refusal, its justification, and its unexplored alternatives.

This extension closes a small loophole: the space of *experiments that could have been run but were not*, together with the decision process that ruled them out. In practice, the preregistration record now contains `unexplored_alternatives` alongside the primary decision basis, so future readers can reconstruct not just what the analyst did but what she chose not to do and why.

1.6.4 5.4 Relation to existing literature

Preregistered rejection is related to but distinct from:

- **Pre-registration registries** (OSF, AsPredicted, PRISMA): these register *when* and *how* an experiment is run; they do not register *refusal to run*;
- **Retraction and correction**: these apply after publication; preregistered rejection applies before execution;
- **Replication registries**: these register *replications*; preregistered rejection registers *non-executions*;
- **Pre-registration of null hypotheses**: these register the *intent* to test null hypotheses; preregistered rejection registers the *decision not to test*.

To the best of our knowledge, preregistered rejection as a formal category is not covered by any of these frameworks.

1.6.5 5.5 Machine enforcement and the anti-rationalization property

Gate 47 enforces preregistered rejection integrity by:

- Independent recomputation from raw samples (not the analyst's summary);
- Explicit declaration of `criteria_modified_after_results = False`;
- Population of `unexplored_alternatives` as a non-trivial field;
- Recording of `rule_underpowered` when applicable (preventing silent threshold changes).

This anti-rationalization property is what makes preregistered rejection *credible*. Without machine enforcement, an analyst could easily rationalize a refusal post-hoc; with it, the refusal must be justified before execution and any post-hoc adjustment is detected.

1.7 6. Conclusion

We proposed **preregistered rejection** as a third category of preregistered output, alongside preregistered success and failure. We formalized it as a tuple that includes the proposed experiment, the pre-execution decision, an independent justification, unexplored alternatives, and a lift condition. We described Gate 47, a machine-enforced integrity checker that verifies the anti-rationalization property. We illustrated the category with PR-2b — a case in which the pre-execution analysis proved analytically (and confirmed empirically at |corr| = 0.902) that the proposed substitute metric was redundant with the intervention.

The interpretive value of preregistered rejection is threefold:

1. **Documentation of decisions not taken** — the space of refused experiments is a scientific record in its own right;
2. **Boundary identification** — preregistered rejection pinpoints substrate limits that cannot be resolved by more data;
3. **Anti-rationalization** — machine enforcement prevents post-hoc rationalization of why an experiment was not run.

We argue that preregistered rejection is scientifically *more informative* than preregistered failure at any sample size, when the failure mode is a substrate limit rather than a sample-size limit. It tells future researchers not to run more of the same — a message that preregistered failure alone cannot convey.

Future work: (i) apply preregistered rejection to a second domain to test its generality; (ii) develop a public registry for preregistered rejections; (iii) investigate whether other preregistered categories (e.g., preregistered abandonment, preregistered scope change) deserve formal treatment.

1.8 Data and code availability

- Preregistered rejection record: `research_claims.json.preregistered.PR2b`
- Pre-execution analysis script: `scripts/f4b_metric_substitution_check.py`

- Integrity checker (Gate 47): `scripts/check_metric_substitution.py`
- Full preregistration protocol: `docs/PREREGISTRATION_PROTOCOL_v1.md#PR-2b`
- Main paper (§4.8 discusses PR-2b in context): `docs/arxiv_paper.md`

Repository: `lm203688/flybrain-connectome` (MIT, public since 2026-10-18, pinned at v2.98.1 with 60/60 gates PASS). The 5.63 GB raw connectome is not redistributed; obtain MaleCNS v1.0 from the original publication.

1.9 Acknowledgments

The preregistration protocol (R-0.1 time-ordering, R-0.2 machine fixation, R-0.3 one-shot verification, R-0.4 failed-experiment publication) and its extension to preregistered rejection (R-0.4′ not-run-experiment publication) were developed as part of the `flybrain-connectome` project. The case study draws on a series of preregistered experiments (PR-1, PR-2, PR-3, PR-4) conducted over the period 2026-10-09 to 2026-10-15.

1.10 References

1.10.1 Primary framework

1. **Preregistration Protocol v1** (`docs/PREREGISTRATION_PROTOCOL_v1.md`) — this project's preregistration standard. R-0.1 time-ordering, R-0.2 machine fixation, R-0.3 one-shot verification, R-0.4 failed-experiment publication. Extended here to R-0.4′ not-run-experiment publication.
2. **Gate 47** (`scripts/check_metric_substitution.py`) — machine-enforced preregistered rejection integrity. Independent recomputation from raw samples, not analyst summaries.

1.10.2 Preregistration infrastructure

3. **The Open Science Framework (OSF)** — osf.io. Preregistration registry and negative-result repository.
4. **AsPredicted** — aspredicted.org. Preregistration service for social and behavioral research.
5. **ClinicalTrials.gov** — clinicaltrials.gov. The US National Library of Medicine's registry for clinical trials, the largest preregistration system in existence.

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1.10.8 This project's prior work

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33. **MaleCNS v1.0** — whole-brain connectome of the male *Drosophila*, *Cell*, 2026 (CC-BY). Source data for all quantitative anchors in this project.

34. **Four Disciplines for Self-Verification in Computational Experiments** — companion paper, `docs/S6S9_SELFVERIFICATION_DISCIPLINES_PAPER.md`. The sibling discipline of preregistered refusal.

Preprint — under review. Contact: `lm203688@ ...`